Wei Shen | Senior ML Research Scientist

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Google Scholar (1,900+ citations, h-index 20)

Professional Summary

Tech Lead & Senior ML Research Scientist advancing **frontier reasoning systems** and **long-horizon agentic AI**. Shipped **Skywork-R1V3-38B** (SOTA 76.0 MMMU, 3.2k GitHub stars) and production multimodal agents outperforming Manus. 1,900+ citations (h-index 20), NeurIPS Best Paper, ICLR Spotlight. Expert in scalable post-training, GRPO, anti-reward-gaming, and building reliable reasoning models that push the boundaries of AI.

Experience

Skywork AI Inc. **Beijing, China**

Senior ML Research Scientist (RL/Multimodal Team) *Mar 2025 – Present*

- **Agentic RL Post-training (Tech Lead):** Led RL post-training for long-horizon agentic capabilities on general/coding tasks. Achieved strong gains on **PinchBench, Claw-Eval, SWE-Bench Pro, Terminal-Bench 2.0** via reward shaping and anti-reward-gaming. Deployed production browser + multimodal search agent surpassing Manus in efficiency.
- **Skywork-R1V3-38B (Tech Lead):** Spearheaded July 2025 release achieving **SOTA 76.0 MMMU** (64.3% → 76.0%) via **GRPO + curriculum learning**. Open-sourced with **3,162 GitHub stars**, matching entry-level human multimodal reasoning.

Baichuan Inc. **Beijing, China**

Machine Learning Research Scientist (RL) *May 2024 – Mar 2025*

- **Clinic Baichuan Foundation Model (Core Contributor):** Built first capable medical reasoning model reaching **OpenAI o1-mini level**. Delivered **+20%** over base SFT on clinical benchmarks via rule-based RL + reward model; competitive with DeepSeek-R1 on **HealthBench**.
- **Baichuan-4 Post-training:** Boosted coding via **Hybrid RL** (RLHF + RLVR). Achieved **5–10%** gains on MBPP, HumanEval – enabled real-world adoption.
- **Mentorship:** Mentored interns on Reward Modeling; work accepted at **ICLR** and **ACL**.

ByteDance Inc. AI Lab (Seed) **Beijing, China**

Research Intern @ Responsible AI *Aug 2023 – Mar 2024*

- **Robust RLHF:** Investigated RLHF dynamics under noisy rewards; developed stabilization methods for reliable post-training.
- **Novel PPO Formulation:** Designed group-relative PPO reward improving performance by **15%**. Architecturally anticipated **GRPO/ReMax-style** methods before 2025 public release.

Fudan University **Shanghai, China**

Research Intern @ CISL Lab *May 2021 – Jul 2021*

- **ML for EDA:** Applied Graph Neural Networks + RL to routing and placement optimization in chip design automation.

Education

Fudan University **Shanghai, China**

M.S. Computer Science @ Fudan NLP Lab *Sept 2021 – Jan 2024*

Advisor: Prof. Xuanjing Huang, Prof. Xipeng Qiu, Prof. Tao Gui

Research: LLM Alignment, RLHF, ML Robustness

Huazhong University of Science and Technology **Wuhan, China**

B.S. Computer Science *Sept 2016 – May 2020*

Open Source Projects

Tech Lead / Core Contributor
Skywork-OR1 & R1V3 *2025*

- Contributed to **Skywork-OR1-32B** (82.2 AIME24, 73.3 AIME25 — surpassing DeepSeek-R1) and led **Skywork-R1V3-38B** (76.0 MMMU SOTA, 3.2k stars). Major advances in scalable reasoning via multi-stage RL and curriculum learning.
- Led ablation studies delivering **+15%** average gains on math/coding benchmarks; validated curriculum + GRPO pipelines now used for frontier reasoning models.

Core Contributor
MOSS-RLHF *May 2023 – Jul 2023*

- **Pioneered** China's first open-source RLHF framework (**1.4k GitHub stars**), releasing production-ready pipeline foundational to OpenRLHF + VeRL used in frontier systems.
- Designed optimized PPO kernels and stable distributed training enabling stronger, reliable reasoning alignment.

Contributor
Skywork-Reward v2 *Jul 2025*

- Contributed to SOTA universal reward model series (0.6B–8B) trained on 26M preference pairs. 8B surpasses all prior open-source RMs; 0.6B nearly matches previous 27B SOTA — critical for scalable post-training of reasoning models.

Selected Publications

1. **Improving Reinforcement Learning from Human Feedback Using Contrastive Rewards**
   *Wei Shen**, Xiaoying Zhang*, Yuanshun Yao, Rui Zheng, Hongyi Guo, Yang Liu
   arXiv preprint, Jan 2024
2. **Loose Lips Sink Ships: Mitigating Length Bias in RLHF**
   *Wei Shen**, Rui Zheng*, Wenyu Zhan, Jun Zhao, Shihan Dou, Tao Gui, Qi Zhang, Xuanjing Huang
   EMNLP 2024
3. **Skywork-R1V3 Technical Report**
   *Wei Shen**, Jiangbo Pei, Yi Peng, Xuchen Song, Yang Liu, Jian Peng, Haofeng Sun, Yunzhuo Hao, Peiyu Wang, Jianhao Zhang, Yahui Zhou
   Technical Report, Jul 2025
4. **Secrets of RLHF in Large Language Models Part I: PPO**
   Rui Zheng, Shihan Dou, Songyang Gao, Yuan Hua, *Wei Shen*, Binghai Wang, Yan Liu, ... Xuanjing Huang
   Best Paper @ NeurIPS 2023 Instruction Workshop
5. **Improving Generalization of Alignment with Human Preferences through Group Invariant Learning**
   Rui Zheng*, *Wei Shen**, Yuan Hua, Wenbin Lai, Shihan Dou, Yuhao Zhou, Zhiheng Xi, Xiao Wang, Haoran Huang, Tao Gui, Qi Zhang, Xuanjing Huang
   Spotlight @ ICLR 2024
6. **Overcoming Reward Overoptimization via Adversarial Policy Optimization with Lightweight Uncertainty Estimation**
   Xiaoying Zhang*, Jean-Francois Ton*, *Wei Shen*, Hongning Wang, Yang Liu
   Spotlight @ NeurIPS 2024
7. **Linear Alignment: A Closed-form Solution for Aligning Human Preferences without Tuning and Feedback**
   Songyang Gao*, Qiming Ge*, *Wei Shen*, Shihan Dou, Junjie Ye, Xiao Wang, Rui Zheng, Yicheng Zou, Zhi Chen, Hang Yan, Qi Zhang, Dahua Lin
   ICML 2024

Technologies & Skills

Expertise: Frontier Reasoning Models, Long-Horizon Agentic RL, Scalable Post-training, GRPO/PPO, Anti-Reward-Gaming, Reward Modeling, Multimodal Reasoning, Curriculum Learning, Reliable Alignment at Scale

Languages: Python, C++, Java, LaTeX

Frameworks & Tools: PyTorch, DeepSpeed, Megatron-LM, vLLM, TRL, verl, OpenRLHF, Weights & Biases, Docker, Linux, Git, Hugging Face

Natural Languages: Mandarin (Native), English (Professional)